

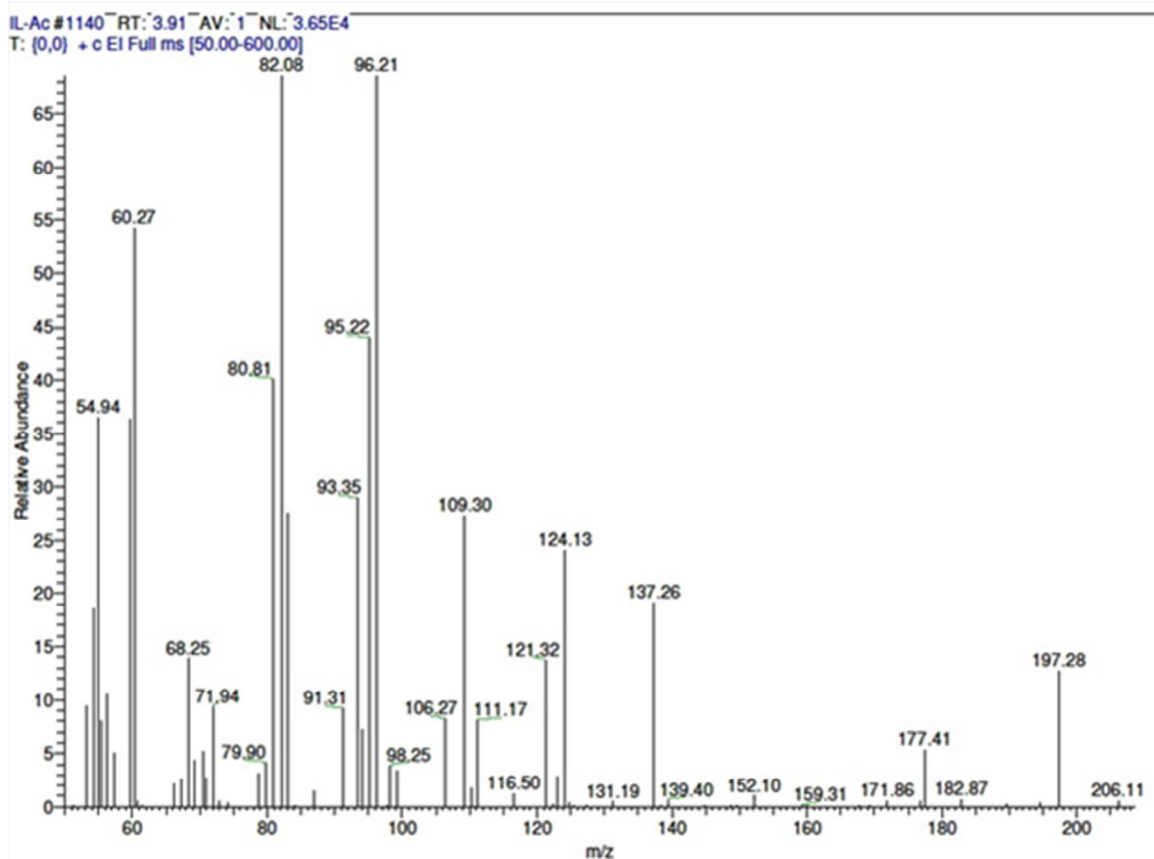


Figure S1. EI-MS spectrum of [Bmim][OAc]

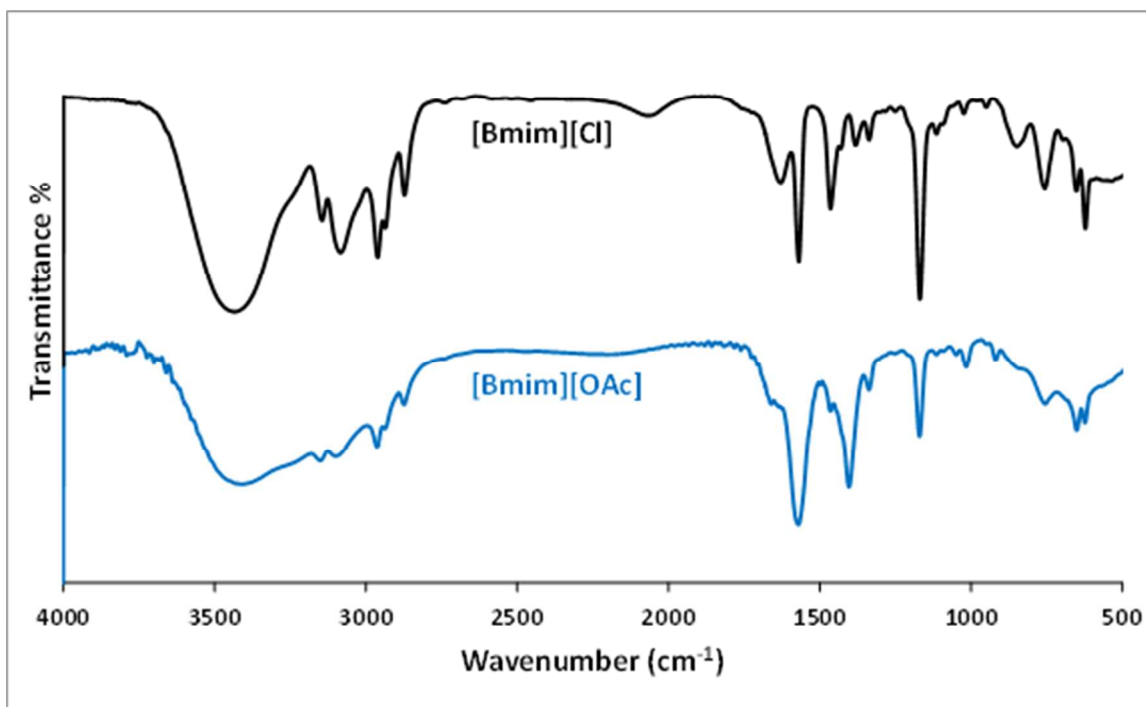


Figure S2. FTIR spectra of [Bmim][OAc] and [Bmim][Cl]

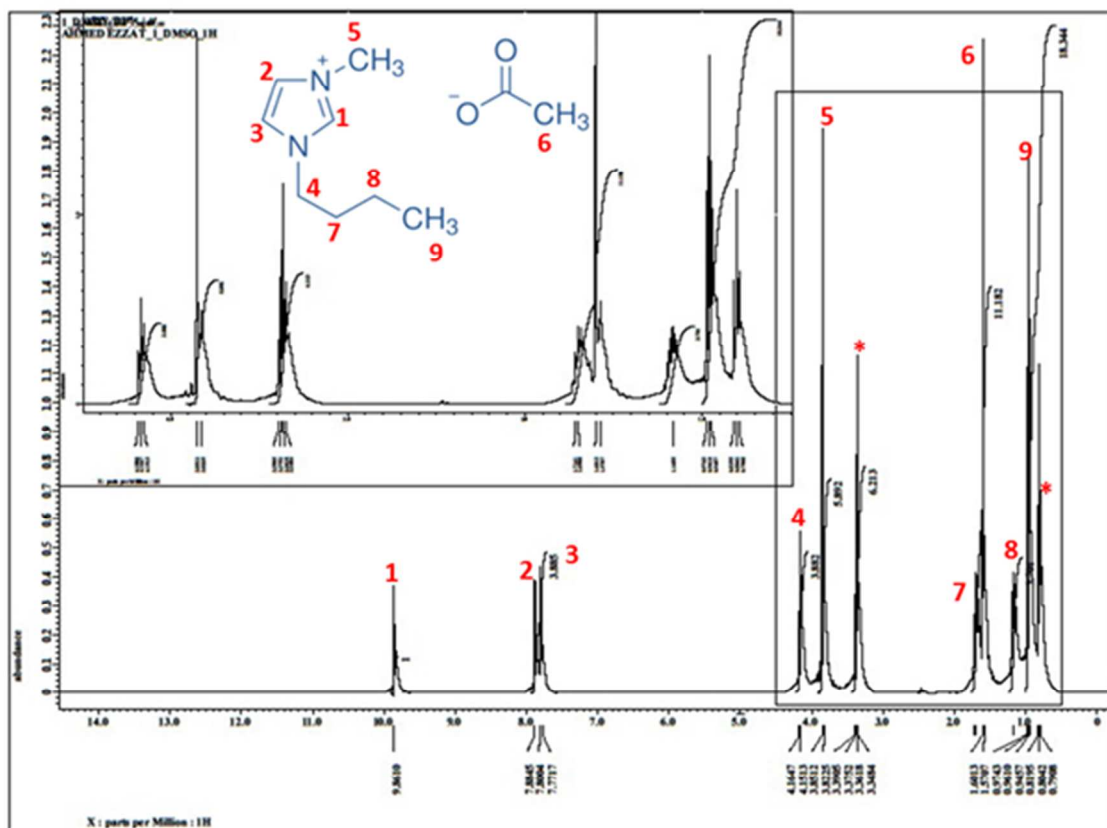


Fig. S3. ^1H NMR spectrum of [Bmim][OAc], where [*] is undefined peak that indicate the presence of contaminants

Table S1Catalytic effect of different ionic liquids on the glycolysis of PET. ^a

Ionic liquids	Conversion of PET (%)	Selectivity of BHET (%)	Yield of BHET (%)
-	0	0	0
[Bmim][Br]	0	0	0
[Bmim][Cl]	0	0	0
[Bmim][OAc]	75.1%	61.9%	46.4%

^a Reaction conditions: PET (2.0 g), EG (20 g), ionic liquid (1.0 g), 1 atm, 180 °C, 2 h.

Table S2Effect of the amount of [Bmim][OAc] on the glycolysis of PET. ^a

Catalyst weight (g)	Conversion (%)	Selectivity (%)	Yield (%)
0.125	14.8	38.3	5.6
0.25	37.95	45	20.3
0.5	48.5	49.87	24.1
1	75.1	61.9	46.4
1.5	55.05	10.29	5.6
2	44.8	4.2	1.8
2.5	42.5	3.2	0.9
3	41.5	0.7	0.29

^a Reaction conditions: PET (2 g), EG (20 g), 1 atm, 180 °C, 2 h.

Table S3Effect of the reaction temperature on the glycolysis of PET.^a

Temperature (°C)	Conversion (%)	Selectivity (%)	Yield (%)
150	11	0	0
170	37	22	8.14
180	75.1	61.9	46.4
190	100	41	41

^a Reaction conditions: PET (2 g), EG (20 g), [Bmim][OAc] (1.0 g), 1 atm, 2 h.

Table S4Effect of the reaction time on the glycolysis of PET. ^a

Reaction Time (h)	Conversion (%)	Selectivity (%)	Yield (%)
1	64	24.8	15.8
2	100	41	41
3	100	53.5	53.5
4	100	44.9	44.9

^a Reaction conditions: PET (2.0 g), EG (20 g), [Bmim][OAc] (1.0 g), 1 atm, 190 °C.

Table S5Effect of the ethylene glycol amount on the glycolysis of PET. ^a

EG amount (g)	Conversion (%)	Selectivity (%)	Yield (%)
5	100	36.6	36.6
10	100	44.2	44.2
20	100	53.5	53.5
25	100	27.5	27.5

^a Reaction conditions: PET (2 g), [Bmim][OAc] (1.0 g) 1 atm, 190 °C, 3 h.

Table S6Effect of the PET amount on the glycolysis of PET. ^a

PET amount (g)	Conversion (%)	Selectivity (%)	Yield (%)
1	100	35.5	35.5
2	100	53.5	53.5
3	100	58.21	58.21
5	100	50.6	50.6
7	91	28.6	26

^a Reaction conditions: EG (20 g), [Bmim][OAc] (1.0 g), 1 atm, 190 °C, 3 h.

Table S7Recycling of the residual EG and [Bmim][OAc].^a

Recycle time	Conversion of PET (%)	Selectivity of BHET (%)	Yield of BHET (%)
0	100%	58.2%	58.2%
1	100%	58.3%	58.3%
2	100%	58.5%	58.5%
3	100%	58.5%	58.5%
4	100%	58.2%	58.2%
5	100%	58.1%	58.1%
6	100%	58%	58%

^a Reaction conditions: PET (3.0 g), EG (20 g), [Bmim][OAc] (1.0 g), 1 atm, 190°C, 3 h.

Table S8Linear regression results for the data in **Figure 3**.

Reaction temperature (°C)	Regressive equation	Linear correlative coefficient
150	$y = 0.0433 + 0.2x$	0.9967
170	$y = 0.26 + 0.338x$	0.9827
180	$y = 0.133 + 0.5123x$	0.979
190	$y = 0.315 + 0.884x$	0.982